

Quantum Information Theory

Lesson 3: Guessing the Winner

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Contents

- ▶ Guessing Entropy
- ▶ Majorization
- ▶ Classical Channels
- ▶ Lorenz Curve
- ▶ Schur Convexity
- ▶ (h, ϕ) -entropies
- ▶ Massey Inequality

Dictionary Attacks

Let's say you packaged a keylogger in some cheating software. Using this, you get a list of account names and passwords. You want to gain access to as many accounts as possible. One strategy is to hack your own clients (bad for business). The better strategy is to leverage this database to attack others. We want to estimate the number of login attempts required.

Note that a login box only allows a single password at a time. Additionally, it only provides a yes-no answer. So you can't use binary search. You **must** use linear search.

As in AI, the trick is to try the most probable passwords first. Note that our database gives us a density estimate for free. All we have to do is count how many uses each password has. Top gamer passwords cca. 2007: shadow, qwerty, pokemon. Top gamer passwords cca. 2010: shad0w, p0kemon, asdf1234. Default IT passwords cca. 2023: P@ssw0rd, Schimbaparola1!. People are not very original.

Guessing Entropy

Consider a race with n horses. We want to call the winning horse,

$$W \sim \begin{pmatrix} 1 & 2 & \dots & n \\ p_1 & p_2 & \dots & p_n \end{pmatrix}.$$

The optimal guessing strategy is to go in most-likely order.
Let σ be the permutation that sorts the p s decreasingly i.e.

$$W \sim \begin{pmatrix} \sigma(1) & \sigma(2) & \dots & \sigma(n) \\ p_{\sigma(1)} & p_{\sigma(2)} & \dots & p_{\sigma(n)} \end{pmatrix},$$

such that $p_{\sigma(i)} \geq p_{\sigma(j)}$ for all $i < j$.

We thus have probability $p_{\sigma(g)}$ to stop after exactly g guesses.

$$G \sim \begin{pmatrix} 1 & 2 & \dots & n \\ p_{\sigma(1)} & p_{\sigma(2)} & \dots & p_{\sigma(n)} \end{pmatrix},$$

for an average of $\mathbb{E}[G] = \sum_i i p_{\sigma(i)}$, known as **guessing entropy**.

Guessing Entropy - Example

Let's say we are calling a horse race with 6 horses,

$$W \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.1 & 0.2 & 0.05 & 0.15 & 0.3 & 0.2 \end{pmatrix}.$$

To obtain the number of guesses we sort decreasingly,

$$G \sim \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 0.3 & 0.2 & 0.2 & 0.15 & 0.1 & 0.05 \end{pmatrix},$$

so the average number of guesses we need to make is

$$\begin{aligned} \mathbb{E}[G] &= 1 \times 0.3 + 2 \times 0.2 + 3 \times 0.2 + 4 \times 0.15 + 5 \times 0.1 + 6 \times 0.05 \\ &= 0.3 + 0.4 + 0.6 + 0.6 + 0.5 + 0.3 = 2.7, \end{aligned}$$

noting that even though the probabilities were sorted decreasingly, the terms of the sum are not.

Guessing Entropy - Fair Example

Let's say we are calling a horse race with 6 horses,

$$W \sim \begin{pmatrix} A & B & C & D & E & F \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \end{pmatrix}.$$

To obtain the number of guesses we sort decreasingly,

$$G \sim \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \end{pmatrix},$$

so the average number of guesses we need to make is

$$\begin{aligned} \mathbb{E}[G] &= 1 \times 1/6 + 2 \times 1/6 + 3 \times 1/6 + 4 \times 1/6 + 5 \times 1/6 + 6 \times 1/6 \\ &= \frac{6(6+1)}{2} \times 1/6 = 3.5, \end{aligned}$$

which corresponds to the intuition that a fair race is harder to call.

Guessing Entropy - Unfair Example

Let's say we are calling a horse race with 6 horses,

$$W \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.01 & 0.01 & 0.01 & 0.95 & 0.01 & 0.01 \end{pmatrix}.$$

To obtain the number of guesses we sort decreasingly,

$$G \sim \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 0.95 & 0.01 & 0.01 & 0.01 & 0.01 & 0.01 \end{pmatrix},$$

so the average number of guesses we need to make is

$$\begin{aligned} \mathbb{E}[G] &= 1 \times 0.95 + 2 \times 0.01 + 3 \times 0.01 + 4 \times 0.01 + 5 \times 0.01 + 6 \times 0.01 \\ &= 0.95 + \left(\frac{6(6+1)}{2} - \frac{1(1+1)}{2} \right) \times 0.01 = 1.15, \end{aligned}$$

which shows that an unfair race is very predictable and no fun.

Guessing Entropy - Example

Let's say we are calling a horse race with 6 horses,

$$W \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.1 & 0.2 & 0.02 & 0.2 & 0.28 & 0.2 \end{pmatrix}.$$

To obtain the number of guesses we sort decreasingly,

$$G \sim \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 0.28 & 0.2 & 0.2 & 0.2 & 0.1 & 0.02 \end{pmatrix},$$

so the average number of guesses we need to make is

$$\begin{aligned} \mathbb{E}[G] &= 1 \times 0.28 + 2 \times 0.2 + 3 \times 0.2 + 4 \times 0.2 + 5 \times 0.1 + 6 \times 0.02 \\ &= 0.28 + 0.4 + 0.6 + 0.8 + 0.5 + 0.12 = 2.7, \end{aligned}$$

just as hard to call as the first race even if the numbers were different, to the point that one horse is almost out of it.

Live Uncertainty Estimates

We have seen how some races are more uncertain than others. Let's consider how perceived fairness changes with news. Initially,

$$W \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.1 & 0.2 & 0.05 & 0.15 & 0.3 & 0.2 \end{pmatrix}; \quad \mathbb{E}[G(W)] = 2.7.$$

Consider that two of the horses were recently in another race. We would expect that the recent winner's probability increase.

For example, if horse D wins against the favourite, E ,

$$W' \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.1 & 0.2 & 0.05 & 0.2 & 0.25 & 0.2 \end{pmatrix}; \quad \mathbb{E}[G(W')] = 2.85,$$

that could have a Robin-Hood effect increasing $\mathbb{E}[G]$.

Now, if debutante horse C loses against the favourite, E ,

$$W'' \sim \begin{pmatrix} A & B & C & D & E & F \\ 0.1 & 0.2 & 0.02 & 0.2 & 0.28 & 0.2 \end{pmatrix}; \quad \mathbb{E}[G(W'')] = 2.7,$$

that could have a reverse Robin-Hood effect decreasing $\mathbb{E}[G]$.

Majorization

It is now convenient to just look at the sorted probability vectors:

$$\mathbf{p} = [0.3 \quad 0.2 \quad 0.2 \quad 0.15 \quad 0.1 \quad 0.05] ; \quad \mathbb{E} [G(\mathbf{p})] = 2.7;$$

$$\mathbf{p}' = [0.25 \quad 0.2 \quad 0.2 \quad 0.2 \quad 0.1 \quad 0.05] ; \quad \mathbb{E} [G(\mathbf{p}')] = 2.85;$$

$$\mathbf{p}'' = [0.28 \quad 0.2 \quad 0.2 \quad 0.2 \quad 0.1 \quad 0.02] ; \quad \mathbb{E} [G(\mathbf{p}'')] = 2.7.$$

In this form, there is no visible pattern. But look at partial sums:

$$\underline{\mathbf{p}} = [0.3 \quad 0.5 \quad 0.7 \quad 0.85 \quad 0.95 \quad 1] ; \quad \mathbb{E} [G(\mathbf{p})] = 2.7;$$

$$\underline{\mathbf{p}'} = [0.25 \quad 0.45 \quad 0.65 \quad 0.85 \quad 0.95 \quad 1] ; \quad \mathbb{E} [G(\mathbf{p}')] = 2.85;$$

$$\underline{\mathbf{p}''} = [0.28 \quad 0.48 \quad 0.68 \quad 0.88 \quad 0.98 \quad 1] ; \quad \mathbb{E} [G(\mathbf{p}'')] = 2.7.$$

Notice how the partial sums in the middle vector are always lower than the partial sums in the other two. This is called **majorization**.

$$\mathbf{p} \succeq \mathbf{q} \Leftrightarrow \sum_{i=1}^k p_i^{\downarrow} \geq \sum_{i=1}^k q_i^{\downarrow} \quad \forall k,$$

where $p_i^{\downarrow}, q_i^{\downarrow}$ denote the i^{th} largest entry in $\mathbf{p}, \mathbf{q}$.

Majorization

$$\mathbf{p} = [0.3 \quad 0.2 \quad 0.2 \quad 0.15 \quad 0.1 \quad 0.05];$$

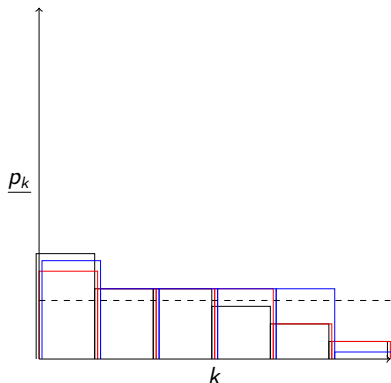
$$\mathbb{E}[G(\mathbf{p})] = 2.7$$

$$\mathbf{p}' = [0.25 \quad 0.2 \quad 0.2 \quad 0.2 \quad 0.1 \quad 0.05];$$

$$\mathbb{E}[G(\mathbf{p}')] = 2.85$$

$$\mathbf{p}'' = [0.28 \quad 0.2 \quad 0.2 \quad 0.2 \quad 0.1 \quad 0.02];$$

$$\mathbb{E}[G(\mathbf{p}'')] = 2.7$$



Majorization

$$\underline{\mathbf{p}} = [0.3 \quad 0.5 \quad 0.7 \quad 0.85 \quad 0.95 \quad 1];$$

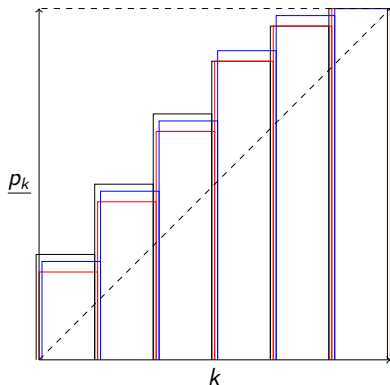
$$\mathbb{E}[G(\mathbf{p})] = 2.7$$

$$\underline{\mathbf{p}}' = [0.25 \quad 0.45 \quad 0.65 \quad 0.85 \quad 0.95 \quad 1];$$

$$\mathbb{E}[G(\mathbf{p}')] = 2.85$$

$$\underline{\mathbf{p}}'' = [0.28 \quad 0.48 \quad 0.68 \quad 0.88 \quad 0.98 \quad 1];$$

$$\mathbb{E}[G(\mathbf{p}'')] = 2.7$$



Majorization

Let $\mathbf{p}, \mathbf{q} \in \mathbb{R}^n$ be probabilistic. The following are equivalent:

1. $\mathbf{p} \succeq \mathbf{q}$;
2. $\mathbf{q} = \mathbf{D}\mathbf{p}$, where $\mathbf{D}$ is a doubly-stochastic matrix (i.e. its rows and columns sum to 1 and all entries are non-negative);
3. $\mathbf{p}$ can be transformed into $\mathbf{q}$ through a **finite** sequence of Robin-Hood operations;
4. $\mathbf{p}$ can be transformed into $\mathbf{q}$ through a (possibly infinitely long) sequence of elementary Robin-Hood operations.

A Robin-Hood operation is a “heist” by which we “take” a fixed amount from the k^{th} largest entry (the “rich”) of $\mathbf{p}$ and “give” it to the j^{th} where $k < j$ (the “poor”) **without rising inequality**.

Elementary Robin-Hood operations take from k giving to $k + 1$. Notice that under such operations, $\mathbb{E}[G]$ necessarily increases, because the “stolen probability” now requires +1 guesses.

Majorization $\Rightarrow$ Robin-Hood

Let's say $\mathbf{p} \succeq \mathbf{q}$ but $\mathbf{p} \neq \mathbf{q}$. We are going to process $\mathbf{p}$ into some $\mathbf{r}$ such that $\mathbf{p} \succeq \mathbf{r} \succeq \mathbf{q}$.

Notice that one of the operations Robin-Hood can perform is swapping. Thus, without loss of generalization we can assume that $\mathbf{p} = \mathbf{p}^\downarrow$ and $\mathbf{q} = \mathbf{q}^\downarrow$. Then, $\mathbf{p} \neq \mathbf{q} \Leftrightarrow \underline{\mathbf{p}}^\downarrow \neq \underline{\mathbf{q}}^\downarrow$ and so we will seek to shrink the sum of absolute differences $\|\underline{\mathbf{p}}^\downarrow - \underline{\mathbf{q}}^\downarrow\|_1$.

Let k be the first mismatch, i.e. s.t. $\sum_{i=1}^k p_i^\downarrow > \sum_{i=1}^k q_i^\downarrow$. Then $p_1^\downarrow, \dots, p_{k-1}^\downarrow$ match $q_1^\downarrow, \dots, q_{k-1}^\downarrow$ and $p_k > q_k$. Since both $\mathbf{p}, \mathbf{q}$ sum to 1, there must be some $j_1, \dots, j_m$ such that $p_{j_i}^\downarrow < q_{j_i}^\downarrow$ and take the first indices such that $p_k^\downarrow - q_k^\downarrow \leq \sum_{i=1}^m q_{j_i}^\downarrow - p_{j_i}^\downarrow$. We will distribute the excess of $p_k^\downarrow$ to the downtrodden $p_{j_i}^\downarrow$, keeping them below $q_{j_i}^\downarrow$.

Notice that all partial sums of $\mathbf{p}$ have been either shrunk or kept intact, but kept above the partial sums of $\mathbf{q}$. Since at every iteration we match one more entry, this process is finite.

Robin-Hood $\Rightarrow$ Bistochastic

Robin-Hood moving ϵ from $p_i^\downarrow$ to $p_j^\downarrow$ means:

$$r_k = p_k \forall k \neq \sigma_p^{-1}(i), \sigma_p^{-1}(j)$$

$$r_{\sigma_p^{-1}(i)} = p_{\sigma_p^{-1}(i)} - \epsilon$$

$$r_{\sigma_p^{-1}(j)} = p_{\sigma_p^{-1}(j)} + \epsilon,$$

and noting that any number $x \in [a, b]$ can be written as $x = \theta a + (1 - \theta)b$ so too we find that there is θ such that

$$r_k = p_k \forall k \neq \sigma_p^{-1}(i), \sigma_p^{-1}(j)$$

$$r_{\sigma_p^{-1}(i)} = \theta p_{\sigma_p^{-1}(i)} + (1 - \theta) p_{\sigma_p^{-1}(j)}$$

$$r_{\sigma_p^{-1}(j)} = (1 - \theta) p_{\sigma_p^{-1}(i)} + \theta p_{\sigma_p^{-1}(j)},$$

a matrix akin to $\mathbf{I}_{n-2} \oplus \begin{bmatrix} \theta & 1 - \theta \\ 1 - \theta & \theta \end{bmatrix}$, which is bistochastic.

This part of the proof is completed by the observation that the product of bistochastic matrices is bistochastic.

Bistochastic $\Rightarrow$ Majorization

The partial sums of $\mathbf{q}^\downarrow$ are linear combinations of the terms of $\mathbf{p}$,

$$\begin{aligned}\sum_{i=1}^k q_i^\downarrow &= \sum_{i=1}^k q_{\sigma_q^{-1}(i)} = \sum_{i=1}^k \sum_{j=1}^n d_{\sigma_q^{-1}(i),j} p_j \\ &= \sum_{j=1}^n \underbrace{\sum_{i=1}^k d_{\sigma_q^{-1}(i),j}}_{c_j \in [0,1]} p_j,\end{aligned}$$

where the sum of coefficients is

$$\sum_{j=1}^n \sum_{i=1}^k d_{\sigma_q^{-1}(i),j} = \sum_{i=1}^k \underbrace{\sum_{j=1}^n d_{\sigma_q^{-1}(i),j}}_1 = k,$$

noting that the sum is maximized when we ascribe maximal coefficients to the bigger probabilities. Yet the biggest coefficient that may arise is 1, so $\sum_{i=1}^k q_i^\downarrow \leq \sum_{i=1}^k p_i^\downarrow$.

Elementary Robin-Hood chains

Prima facie, since S_n is generated by transpositions $(i, i + 1)$, Elementary Robin Hood operations can be used to mediate mass shifting between any two probability indices.

While we may indeed change the indices, these operations do not actually change the order of the actual sorted probabilities: convex combinations of the i^{th} and $(i + 1)^{\text{th}}$ probabilities still yield results that are below the first $i - 1$ and above the last $n - i - 1$.

Consider how we can implement a sequence of elementary Robin-Hood operations enacting

$$\begin{bmatrix} 2/3 & 1/3 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1/3 & 1/3 & 1/3 \end{bmatrix}.$$

That $1/3$ limits how much we can take from $2/3$: in one transfer we can at best reach either $\begin{bmatrix} 1/2 & 1/2 & 0 \end{bmatrix}$ or $\begin{bmatrix} 2/3 & 1/6 & 1/6 \end{bmatrix}$. Next transfer we reach $\begin{bmatrix} 1/2 & 1/4 & 1/4 \end{bmatrix}$ or $\begin{bmatrix} 5/12 & 5/12 & 1/6 \end{bmatrix}$, and so on and while we can at every step strictly improve our objective, we only actually ever reach it in the limit.

Uncertainty vs. Inequality

The concept of Robin-Hood operations comes from economics where cca. 1920 Hugh Dalton postulated that they are **the** operations that should **decrease any measure of inequality**.

In Information Theory we are interested in the opposite: for us these Robin-Hood operations should **increase any measure of uncertainty** such as Guessing Entropy.

As concepts cross over, each field takes ideas from the other. Information Theory results now often use majorization. Meanwhile, economists now also use variations of entropy.

The Lorenz Curve

Working with large data, the previous conditions are intractable. For such scenarios, economists developed a visual technique. Given a random variable X they plot the **Lorenz Curve**.

Let's say x_i is i^{th} smallest value X can take, and $p_i = \Pr[X = x_i]$. Then, first k entries contribute $\underline{p}_k = \sum_{i=1}^k p_i$ to the total $1 = \underline{p}_n$, and $S_k = \sum_{i=1}^k x_i p_i$ to the average $\mathbb{E}[X] = S_n$.

The Lorenz curve plots contribution of the bottom % to the average, i.e. it plots the fraction $L_k = S_k/S_n$ as a function of $\underline{p}_k$. The Lorenz curve also includes the origin, i.e. $\underline{p}_0 = 0, L_0 = 0$.

But what do we do if we only have a probabilistic vector $\mathbf{q}$?

One idea: calculate the Lorenz curve for the histogram of $\mathbf{q}$.

For simplicity, we assume that the entries of $\mathbf{q}$ are distinct.

Then $x_i = q_i^\uparrow$, $p_i = 1/n$, $\underline{p}_k = k/n$, $L_k = \sum_{i=1}^k q_i^\uparrow = 1 - \sum_{i=1}^{n-k} q_i^\downarrow$.

So, if $\mathbf{q} \succeq \mathbf{q}'$ then $L_k \leq L'_k$ i.e. Robin-Hood brings the curve up.

The Gini Coefficient

Note that the Lorenz curve is always convex, because a “richer” 1% will always contribute more to the average than a “poorer” 1%. As Robin-Hood acts, he raises curve toward the diagonal. So, the area between the curve and the diagonal shrinks toward 0. The ratio of this area to the maximal $1/2$ is the **Gini coefficient**.

We now consider our previously engineered Lorenz curve, interpolated linearly. Taking $S_0 = 0$, the area under the curve is

$$\begin{aligned} A &= \frac{1}{n} \sum_{k=0}^{n-1} \frac{L_k + L_{k+1}}{2} = -\frac{1}{2n} + \frac{1}{n} \sum_{k=1}^n L_k \\ &= -\frac{1}{2n} + 1 - \frac{1}{n} \sum_{k=1}^n \sum_{i=1}^{n-k} q_i^\downarrow = \frac{n-1/2}{n} - \frac{1}{n} \sum_{i=1}^n (n-i) q_i^\downarrow \\ &= \frac{\mathbb{E}[G] - 1/2}{n} \implies Gini = 1 - 2A = \frac{n+1}{n} - \frac{2\mathbb{E}[G]}{n}. \end{aligned}$$

The Lorenz Curve

For the numerical variable:

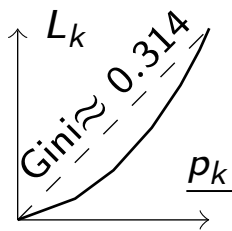
$$G \sim \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 0.3 & 0.2 & 0.2 & 0.15 & 0.1 & 0.05 \end{pmatrix},$$

$$\mathbf{p} = [0.3 \quad 0.2 \quad 0.2 \quad 0.15 \quad 0.1 \quad 0.05]$$

$$\underline{\mathbf{p}} = [0.3 \quad 0.5 \quad 0.7 \quad 0.85 \quad 0.95 \quad 1]$$

$$\mathbf{S} = [0.3 \quad 0.7 \quad 1.3 \quad 1.9 \quad 2.4 \quad 2.7]$$

$$\mathbf{L} \approx [0.111 \quad 0.259 \quad 0.481 \quad 0.703 \quad 0.888 \quad 1]$$



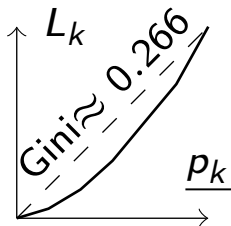
For the probabilities:

$$\mathbf{q}^\uparrow = [0.05 \quad 0.1 \quad 0.15 \quad 0.2 \quad 0.2 \quad 0.3]$$

$$\underline{\mathbf{p}} \approx [0.166 \quad 0.333 \quad 0.5 \quad 0.666 \quad 0.833 \quad 1]$$

$$\mathbf{L} = [0.05 \quad 0.15 \quad 0.3 \quad 0.5 \quad 0.7 \quad 1]$$

$$\mathbb{E}[G] = 2.7 \Rightarrow Gini = \frac{n+1 - 2\mathbb{E}[G]}{n} = 0.2(6)$$

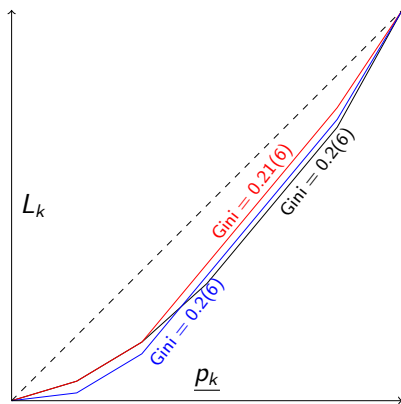


The Lorenz Curve

$$\mathbf{q}_1^\uparrow = [0.05 \quad 0.1 \quad 0.15 \quad 0.2 \quad 0.2 \quad 0.3]; \quad \mathbb{E} [G(\mathbf{q}_1^\uparrow)] = 2.7$$

$$\mathbf{q}_2^\uparrow = [0.05 \quad 0.1 \quad 0.2 \quad 0.2 \quad 0.2 \quad 0.25]; \quad \mathbb{E} [G(\mathbf{q}_2^\uparrow)] = 2.85$$

$$\mathbf{q}_3^\uparrow = [0.02 \quad 0.1 \quad 0.2 \quad 0.2 \quad 0.2 \quad 0.28]; \quad \mathbb{E} [G(\mathbf{q}_3^\uparrow)] = 2.7$$



Schur-Convexity

We now look at recipes for how to build measures of uncertainty. Ideally, such functions would reverse the majorization order. Functions that preserve majorization are known as Schur-convex. Functions that reverse majorization are known as Schur-concave.

$$f \text{ is Schur-convex} \Leftrightarrow f(\mathbf{p}) \geq f(\mathbf{q}) \forall \mathbf{p} \succeq \mathbf{q};$$

$$f \text{ is Schur-concave} \Leftrightarrow f(\mathbf{p}) \leq f(\mathbf{q}) \forall \mathbf{p} \succeq \mathbf{q}.$$

Gini is Schur-convex, while $\mathbb{E}[G]$ is Schur-concave.

Theorem (Schur-Ostrowski Criterion)

If $f \in \mathcal{C}_1(\mathbb{R}^n \rightarrow \mathbb{R})$ is symmetric then f is Schur-convex iff.

$$(x_i - x_j) \left(\frac{\partial f}{\partial x_i} - \frac{\partial f}{\partial x_j} \right) \geq 0 \forall \mathbf{x} \in \mathbb{R}^n.$$

Corollary

If $\phi : [0, 1] \rightarrow \mathbb{R}$ is convex then $f(\mathbf{x}) = \sum_i \phi(x_i)$ is Schur-convex.

(h, ϕ) -entropies

Consider now a finite sequence of functions (h_k, ϕ_k) such that, for every k either h_k is decreasing and ϕ_k is convex or h_k is increasing and ϕ_k is concave. Also $h_k(0) = \phi_k(0) = h_k(\phi_k(1)) = 0$. Then,

$$\mathcal{H}(\mathbf{x}) = \sum_k h_k \left(\sum_{i=1}^n \phi_k(x_i) \right)$$

is Schur-concave and known as a (h, ϕ) -**entropy**.

Examples:

- ▶ $\phi(x) = x^2$ is convex so the **purity** $P(\mathbf{p}) = \sum_i p_i^2$ is Schur-convex;
- ▶ $\phi(x) = x \log x$ is convex so the **Shannon entropy** $H(\mathbf{p}) = -\sum_i p_i \log p_i$ is Schur-concave;
- ▶ If $q > 0$, $\phi(x) = x^q$ is convex iff $q \geq 1$ so the **Tsallis entropy** $S_q(\mathbf{p}) = \frac{1}{q-1}(1 - \sum_i p_i^q)$ is Schur-concave;
- ▶ If $\alpha > 0$, $\phi(x) = x^\alpha$ is convex iff $\alpha \geq 1$ so the **Rényi entropy** $H_\alpha(\mathbf{p}) = \frac{1}{1-\alpha} \log \sum_i p_i^\alpha = \frac{\alpha}{\alpha-1} \log \|\mathbf{p}\|_\alpha$ is Schur-concave;

(h, ϕ) -entropies

Consider now a finite sequence of functions (h_k, ϕ_k) such that, for every k either h_k is decreasing and ϕ_k is convex or h_k is increasing and ϕ_k is concave. Also $h_k(0) = \phi_k(0) = h_k(\phi_k(1)) = 0$. Then,

$$\mathcal{H}(\mathbf{x}) = \sum_k h_k \left(\sum_{i=1}^n \phi_k(x_i) \right)$$

is Schur-concave and known as a (h, ϕ) -**entropy**.

Similarly, we have the (h, ϕ) -**divergence**,

$$\mathcal{D}(\mathbf{p} \parallel \mathbf{q}) = - \sum_k h_k \left(\sum_{i=1}^n q_i \phi_k \left(\frac{p_i}{q_i} \right) \right),$$

defined for $\mathbf{p}, \mathbf{q}$ s.t. if q_i is 0, p_i is also 0 i.e. $\text{supp } \mathbf{p} \subseteq \text{supp } \mathbf{q}$.

(h, ϕ) -entropies

Consider now a convex function ϕ and let's look at the function

$$f(x, y) = y\phi(x/y)$$

$$\frac{d}{dx}f(x, y) = \phi'(x/y)$$

$$\frac{d}{dy}f(x, y) = \phi(x/y) - \frac{x}{y}\phi'(x/y)$$

$$\frac{d^2}{dx^2}f(x, y) = \frac{1}{y}\phi''(x/y)$$

$$\frac{d^2}{dydx}f(x, y) = -\frac{x}{y^2}\phi''(x/y)$$

$$\frac{d^2}{dx dy}f(x, y) = -\frac{x}{y^2}\phi''(x/y)$$

$$\frac{d^2}{dy^2}f(x, y) = \frac{x^2}{y^3}\phi''(x/y)$$

$$\mathbf{H}(f) = \underbrace{\begin{vmatrix} \underbrace{1/y}_{>0} & -x/y^2 \\ -x/y^2 & x^2/y^3 \end{vmatrix}}_{=0} \underbrace{\phi''(x/y)}_{\geq 0}$$

$\mathbf{H}(f) \geq 0$ so f is convex i.e. $f\left(\frac{x+\tilde{x}}{2}, \frac{y+\tilde{y}}{2}\right) \leq \frac{1}{2}f(x, y) + \frac{1}{2}f(\tilde{x}, \tilde{y})$.

(h, ϕ) -entropies

Let's now consider any $a, b \geq 0$.

Since for any convex ϕ , the function $f(x, y) = y\phi(x/y)$ is convex,

$$\begin{aligned}\mathcal{D}(a\mathbf{p} + b\tilde{\mathbf{p}} \| a\mathbf{q} + b\tilde{\mathbf{q}}) &= \sum_{i=1}^n (aq_i + b\tilde{q}_i) \phi\left(\frac{ap_i + b\tilde{p}_i}{aq_i + b\tilde{q}_i}\right) \\ &\leq \frac{a}{a+b} \sum_{i=1}^n (a+b)q_i \phi\left(\frac{(a+b)p_i}{(a+b)q_i}\right) + \frac{b}{a+b} \sum_{i=1}^n (a+b)\tilde{q}_i \phi\left(\frac{(a+b)\tilde{p}_i}{(a+b)\tilde{q}_i}\right) \\ &= a \sum_{i=1}^n q_i \phi\left(\frac{p_i}{q_i}\right) + b \sum_{i=1}^n \tilde{q}_i \phi\left(\frac{\tilde{p}_i}{\tilde{q}_i}\right) = a\mathcal{D}(\mathbf{p} \| \mathbf{q}) + b\mathcal{D}(\tilde{\mathbf{p}} \| \tilde{\mathbf{q}}),\end{aligned}$$

so all (h, ϕ) -divergences are **joint-convex** (and not just that).

Note that we did not require $a + b = 1$, e.g. $a = 2, b = 3$ works.

(h, ϕ) -entropies

Now that we know (h, ϕ) -divergences are jointly-convex, let's look at what happens under a stochastic matrix $\mathbf{D} \in \text{Mat}_{m,n}(\mathbb{R})$,

$$\begin{aligned}\mathcal{D}(\mathbf{D}\mathbf{p} \parallel \mathbf{D}\mathbf{q}) &= \sum_{j=1}^m f((\mathbf{D}\mathbf{p})_j, (\mathbf{D}\mathbf{q})_j) = \sum_{j=1}^m f\left(\sum_{i=1}^n D_{j,i} p_i, \sum_{i=1}^n D_{j,i} q_i\right) \\ &\leq \sum_{j=1}^m \sum_{i=1}^n D_{j,i} f(p_i, q_i) = \sum_{i=1}^n \underbrace{\sum_{j=1}^m D_{j,i}}_{=1} f(p_i, q_i) = \mathcal{D}(\mathbf{p} \parallel \mathbf{q}),\end{aligned}$$

so (h, ϕ) -divergences do not increase under stochastic operations. Notice that we did not require $\mathbf{D}$ to be doubly-stochastic.

$\mathcal{D}(\mathbf{D}\mathbf{p} \parallel \mathbf{D}\mathbf{q}) \leq \mathcal{D}(\mathbf{p} \parallel \mathbf{q})$ is called the **data processing inequality**. While the DPI is related to joint-convexity, not every function $\mathcal{D}$ that is joint-convex satisfies the DPI nor *vice versa*.

(h, ϕ) -entropies

Now, let's look at the channel whose matrix is:

$$\mathbf{D} = \begin{bmatrix} p_1 & p_2 & \dots & p_{n-1} & p_n \\ p_2 & p_3 & \dots & p_n & p_1 \\ \vdots & \vdots & & \vdots & \vdots \\ p_n & p_1 & \dots & p_{n-2} & p_{n-1} \end{bmatrix} \Rightarrow \mathbf{D} |0\rangle = \mathbf{p}, \mathbf{D} \frac{\mathbf{1}}{n} = \frac{\mathbf{1}}{n},$$

and the channel whose $\mathbf{D} = \mathbf{1}/n$ which turns all $\mathbf{p}$ into $\frac{\mathbf{1}}{n}$. We get,

$$\mathcal{D} \left(|0\rangle \parallel \frac{\mathbf{1}}{n} \right) \geq \mathcal{D} \left(\mathbf{p} \parallel \frac{\mathbf{1}}{n} \right) \geq \mathcal{D} \left(\frac{\mathbf{1}}{n} \parallel \frac{\mathbf{1}}{n} \right) = 0.$$

Actually, any divergence gives rise to a (h, ϕ) -entropy,

$$\begin{aligned} \tilde{\mathcal{H}}(\mathbf{p}) &= \mathcal{D} \left(|0\rangle \parallel \frac{\mathbf{1}}{n} \right) - \mathcal{D} \left(\mathbf{p} \parallel \frac{\mathbf{1}}{n} \right) \\ &= \underbrace{\mathcal{D} \left(|0\rangle \parallel \frac{\mathbf{1}}{n} \right)}_{h_0 = \text{constant}} + \sum_k h_k \left(\sum_{i=1}^n \underbrace{\frac{\phi_k(np_i)}{n}}_{\tilde{\phi}_k(x) = \phi_k(nx)/n} \right). \end{aligned}$$

Massey Inequality

We have seen many quantifiers today. The main scope of information theory is finding out how such quantifiers relate to one another. We now look at guessing entropy and Shannon entropy.

For example, let's take a random variable X with $\mathbb{E}[G(X)] = g$. Can we find an upper bound for $H(X)$?

$$\max \left\{ - \sum_{i=1}^{\infty} p_i \log p_i \mid \sum_{i=1}^{\infty} i p_i = g, \sum_{i=1}^{\infty} p_i = 1 \right\}$$

Note that we are maximizing a concave function over a convex set (the intersection of two hyperplanes). We use the Lagrangian

$$\mathcal{L}(\mathbf{p}) = - \sum_{i=1}^{\infty} p_i \log p_i + \lambda \left(\sum_{i=1}^{\infty} i p_i - g \right) + \mu \left(\sum_{i=1}^{\infty} p_i - 1 \right),$$

and notice that when it is restricted to our initial convex set, the maximum of the Lagrangian is the answer to our problem.

Massey Inequality

We need to maximize the Lagrangian

$$\mathcal{L}(\mathbf{p}) = - \sum_{i=1}^{\infty} p_i \log p_i + \lambda \left(\sum_{i=1}^{\infty} i p_i - g \right) + \mu \left(\sum_{i=1}^{\infty} p_i - 1 \right),$$

so we set

$$0 = \frac{\partial \mathcal{L}}{\partial p_i} = -\log p_i - \frac{1}{\ln 2} + \lambda i + \mu \Leftrightarrow p_i = 2^{\lambda i + \mu - \log e} = \underbrace{(2^{\lambda + \mu} / e)}_{p_1} \underbrace{(2^{\lambda})}_{1 - p_1}^{i-1}$$

so the Lagrangian is maximized by the geometric distribution with

$$\begin{aligned} g &= \sum_{i=1}^{\infty} i p_i = 1 + \sum_{i=2}^{\infty} (i-1) p_i = 1 + (1 - p_1) \sum_{i=2}^{\infty} (i-1) p_{i-1} \\ &= 1 + (1 - p_1) \sum_{j=1}^{\infty} j p_j = 1 + (1 - p_1) g \Rightarrow p_1 = 1/g. \end{aligned}$$

Massey Inequality

Let's take a random variable X with $\mathbb{E}[G(X)] = g$.

Can we find an upper bound for $H(X)$?

$$\max \left\{ - \sum_{i=1}^{\infty} p_i \log p_i \mid \sum_{i=1}^{\infty} i p_i = g, \sum_{i=1}^{\infty} p_i = 1 \right\}$$

We know that this maximum is given by the distribution

$$p_i = \frac{1}{g} \left(1 - \frac{1}{g}\right)^{i-1} = \frac{1}{g-1} \left(1 - \frac{1}{g}\right)^i,$$

so we only need to calculate

$$\begin{aligned} - \sum_{i=1}^{\infty} p_i \log p_i &= - \sum_{i=1}^{\infty} p_i \log \left(\frac{1}{g-1} \left(1 - \frac{1}{g}\right)^i \right) \\ &= \log(g-1) - \log(1 - 1/g) \sum_{i=1}^{\infty} i p_i \\ &= \log(g-1) - g \log(1 - 1/g). \end{aligned}$$

Massey Inequality

Let's take a random variable X with $\mathbb{E}[G(X)] = g$.
Can we find an upper bound for $H(X)$? We found

$$H(X) \leq \log(g-1) + \underbrace{g \log(1 - 1/g)}_{f(g)},$$

where $f(g)$ is decreasing between $f(1) = 2$ and $f(\infty) = 0$ so, if $\mathbb{E}[G(X)] \geq 1$ (or equivalently if $H(X) \geq 2$) we have

$$H(X) \leq \log(\mathbb{E}[G(X)] - 1) + 2,$$

or equivalently, as the Massey inequality is most often written:

$$\mathbb{E}[G(X)] \geq 1 + 2^{H(X)-2},$$

valid when $H(X) \geq 2$.

Massey Inequality

Let's say we have n independent random variables $X_1, \dots, X_n$.

Let's say we are guessing them all at once so we want to compute

$$\mathbb{E}[G(X_1, X_2, \dots, X_n)] \geq 1 + \frac{1}{4} 2^{H(X_1, X_2, \dots, X_n)} = 1 + \frac{1}{4} 2^{\sum_{i=1}^n H(X_i)},$$

where the lower bound on the right can be computed in parallel while the quantity on the left requires sorting a very large array.

For example, if we have $16 \times$ variables that can each take 256 values, the tuple can take $2^{8 \cdot 16} = 2^{128} \approx 256 \times 10^{12}$ values.

Now, if we want these values to be accurate, we will need more than a double per value i.e. more than $2PB$ to store them.

On the other hand, for the same example, for the entropy we only need to compute 16 sums of 256 elements each.

Information theory research derives relations between quantifiers you need and quantifiers you can measure.

Follow-up Reading

- ▶ J.L. Massey – Guessing and Entropy
Proc. 1994 IEEE Intl. Symposium on Information Theory
- ▶ B.C.Arnold – Majorization and the Lorenz Order: A Brief Introduction
Springer (1987) Lecture Notes in Statistics
- ▶ M.Salicru et al. – Asymptotic distribution of (h, ϕ) -entropies
Communications in Statistics-Theory and Methods 22.7
(1993) pp. 2015-2031
- ▶ L.Pardo et al. – Generalized divergence measures: Information matrices, amount of information, asymptotic distribution, and its applications to test statistical hypotheses
Information sciences 84.3-4 (1995): 181-198.